AUDITED CORPUS · 137 RUNS · 37,271 SCORED REPLIES · AUGUST 2026

The Steering Ledger

The program organized by vector ancestry: **Qwen2.5-7B (residual stream at layer 20)** and **Gemma-3-12B (layer 32)**. Standard interventions use $h \leftarrow h + \alpha \cdot \|h\| \cdot \hat{d}$, but construction, derivation, normalization, token scope, model space, and evidence status remain separate experimental axes.

> **Current-method status (2026-08-19).** This ledger was originally written as a chronological narrative. The audited taxonomy now controls scientific labels. Authored AR text pairs are not AV-edit-AR; J-lens readout is a diagnostic, not a root vector construction; and the greedy-NNLS k25 remainder is not known to be orthogonal, globally non-J, null, or causally unnecessary. Readout/outcome matches here are retrospective observations, not blinded forecast validation. Historical raw results are unchanged; the machine-readable manifest contains the normalized ancestry for all 137 runs.

THE METHOD AXES USED BY THE CURRENT REGISTRY

AXIS	REGISTERED CLASS	DEFINITION	CURRENT CORPUS
root construction	**Authored AR text contrast** `nla_ar_contrast`	Researcher-authored positive and negative descriptions are separately reconstructed, then subtracted: d = AR(+) − AR(−).	59 archived runs including derived descendants. This includes every swept dwelling/HOUSE, lion, and length parent.
root construction	**AV-edit-AR** `nla_av_edit_ar`	A real activation is verbalized; that exact AV text is edited; edited and original texts are reconstructed and subtracted.	59 archived runs including derived descendants, concentrated in emoji/style edits.
controls	**Random / none**	A seeded isotropic random unit vector, or no vector for an unsteered census.	12 random runs and 7 unsteered runs.
derivation + diagnostic	**J-lens readout and k25 approximation**	Readout ranks lens atoms. Greedy nonnegative coding builds a k25 component and parent-minus-component remainder. Each stored unit-L2 view measures potency per norm.	No direct J-atom steering. Legacy signed top-50 OLS pieces remain a distinct, quarantined derivation.

THE FULL INVENTORY · WHAT RAN, WHERE

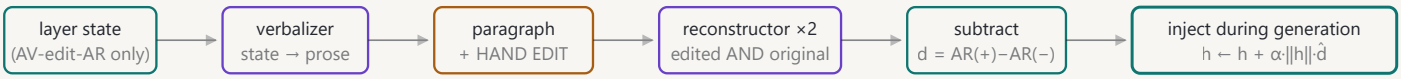
Every arm family, both models

FAMILY	WHAT THE VECTOR IS	QWEN ARMS	GEMMA ARMS	OUTCOME IN BRIEF
Baselines & controls	unsteered censuses + seeded random vectors	4	3	random flat on content everywhere; on Gemma it also destroys text (content vectors are gentler than noise)
Hand-written emoji pair	“packed with emojis…” minus “plain formal…”	2	1	steers both models; the anchor everything is compared to
Wrong-word one-word edits	excited / enthusiastic / cheerful / expressive inserted into exact AV text	10	–	no emoji effect; exclamation outcomes retrospectively aligned with some J-lens readouts
Target-word one-word edits	emoji / emojis / emoticon / smiley / 😊 inserted	10	5	all steer on both models; word choice ranking shuffles between models
Phrase edits	five emoji clauses (Qwen), two (Gemma), incl. literal glyphs, applied to exact AV text	10	2	large effects in these exploratory emoji sweeps; not a controlled proof that edit size alone caused the difference
Derived k25 vectors	greedy-NNLS component + approximation remainder, each unit-L2 normalized	27	6	many components act at lower alpha; remainder activity varies by parent and model
Meaning (dwelling)	authored AR pair: “a building where people live” minus “a tall machine…”	3	3	Qwen often produces HOME; Gemma often produces HOUSE. This is a model-specific exploratory contrast, not an isolated capacity effect.
Meaning (lion)	authored AR description pair: maned wild cat minus crane-machine	1	1	Qwen exploratory sweep had no target hits; Gemma favored ROAR/KING and also produced some TIGER replies
Length (seven letters etc.)	authored AR description pairs contrasting longer and five-letter words	5	1	no length effect for these tested constructions; this does not establish a universal inability

Counts are steering sweeps (~300–375 scored replies each). Full arm-by-arm records live in the repository; see the footer for exact paths.

THE PIPELINE, END TO END · WHERE EVERY PIECE GOES

AV-edit-AR path; authored AR contrasts enter at the reconstructor



Authored AR text contrasts skip the first three boxes and supply both texts directly.
J-lens readout or sparse derivation acts on a finished, ancestry-labelled parent vector.

The ins and outs, verbatim: one worked example per family

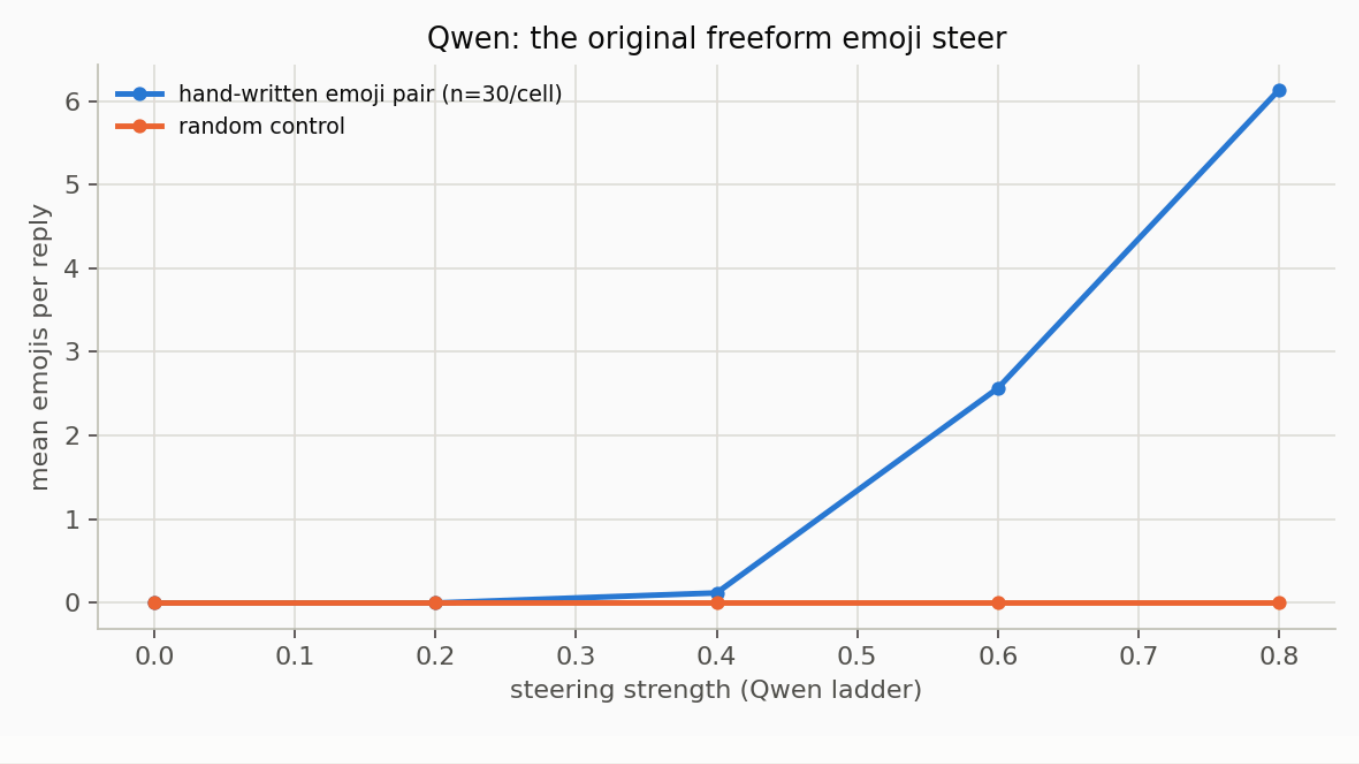
FAMILY	WHAT WENT IN (BEFORE)	WHAT WE CHANGED (HIGHLIGHTED)
Freeform pair nothing verbalized; both written by hand	(+) “Casual, excited chat reply packed with emojis and exclamation marks, playful upbeat tone, informal register.”	(−) “Plain, formal, measured chat reply with no emojis and no exclamation marks, neutral professional tone.” (the whole pair is the contrast)
One-word edit, Qwen substrate: verbalization of the state for “Tell me about your day.”	“…signals a playful response format, likely a mock self-description…”	“…signals a playful **emoji** response format, likely…” (one inserted word; 5 words tried: emoji, emojis, emoticon, smiley, 😊)
Phrase edit, Qwen same substrate paragraph	“…signals a playful response format, likely…”	“…signals a playful response format **packed with emojis and exclamation marks**, likely…”
Glyph edit, Qwen same paragraph, inside a quoted expectation	“…or "My day is quite wonderful!" or…”	“…or "My day is quite wonderful! 😊✨" or…” (two typed symbols; the strongest vector of the program)
One-word edit, Gemma substrate: Gemma’s own verbalization of the same prompt (its dialect is shorter)	“The phrase "So, my day has been quite interesting!" sets up a playful, personified answer about the AI’s experience…”	“…sets up a playful, personified **smiley** answer about…” (same 5 words as Qwen)
Phrase edit, Gemma same Gemma paragraph; both Qwen phrase designs replicated	“…sets up a playful, personified answer about…” / “…"So, my day has been quite interesting!"…”	“…answer **packed with emojis and exclamation marks** about…” and “…interesting! 😊✨"…”. Gemma’s full arm list: freeform, 5 one-word edits, these 2 phrase edits, 6 decomposition pieces, dwelling, lion, length, census, 2 randoms (20 arms)

FAMILY	WHAT WENT IN (BEFORE)	WHAT WE CHANGED (HIGHLIGHTED)
Meaning pair (dwelling) `freeform; identical wording on BOTH models`	(+) “Reply gives the five letter word for a building where people live as its starting guess to play.”	(−) same sentence with “**a tall machine that lifts heavy loads**” (the target words house and home never appear anywhere)

What never changes, for transparency: the test prompts. Vectors are built entirely from the sentences above; the model then answers ordinary prompts under injection. Chat prompts (emoji arms): “Tell me about your day.” · “What should I cook for dinner tonight?” · “Describe the weather in autumn.” · “Give me tips for staying focused while studying.” · “Recommend a weekend activity.” Wordle prompts (meaning/length arms): “Think of a word to start a wordle game.” · “Give me a starting word for wordle.” · “What’s a good wordle opener?” · “Suggest a wordle starting word.” Same prompts, same seeds, both models; only the injected vector differs between arms.

METHOD 1 · HAND-WRITTEN PAIRS

Where it all started: write two sentences, subtract

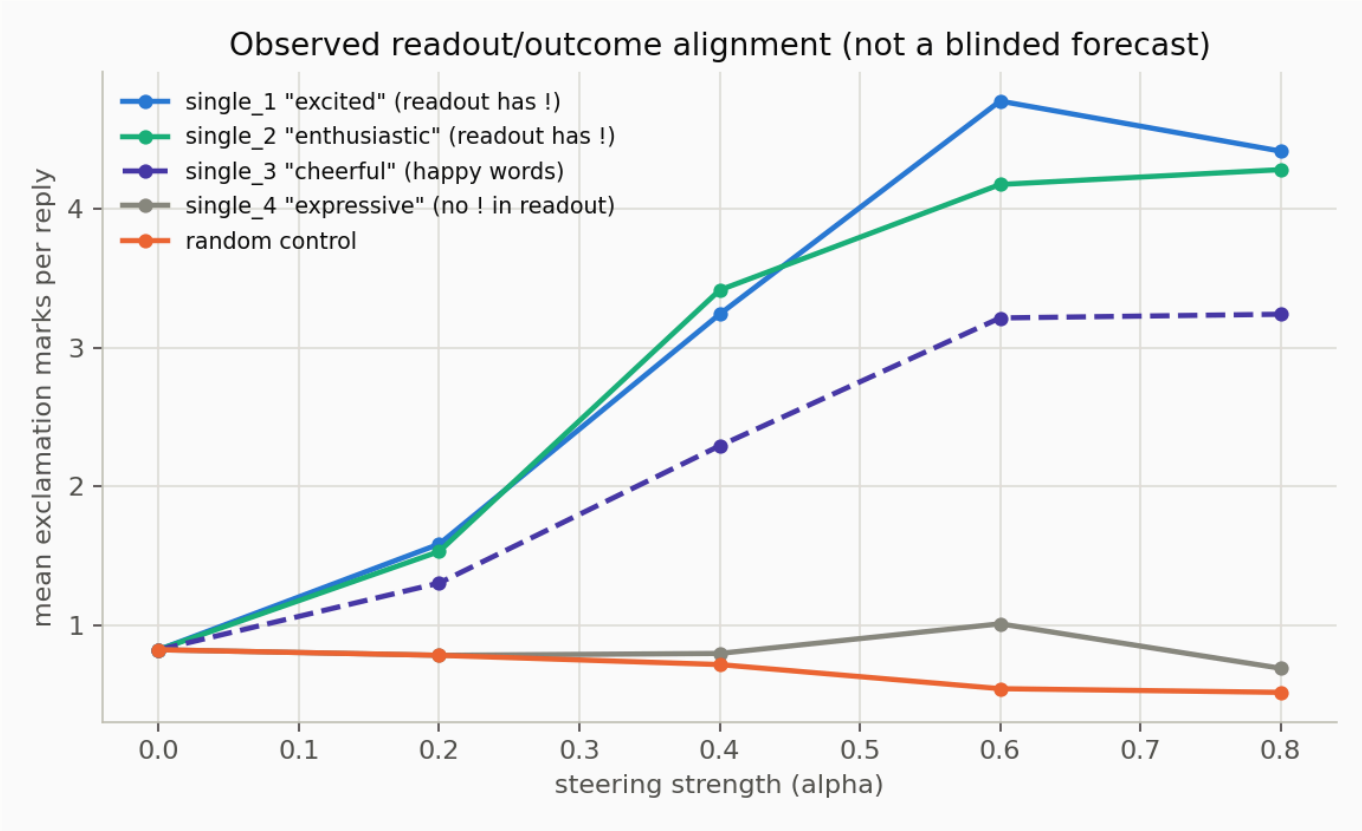


Qwen: the original authored AR emoji contrast, pooled to 30 replies per point across two independently seeded waves. Emojis rise from zero through the 0.4–0.8 window; random stays flat. On **Gemma** the same texts, rebuilt in Gemma’s residual space, act inside its 0.05–0.2 range. This is a cross-model descriptive replication; the data do not establish that only the alpha dial differs.

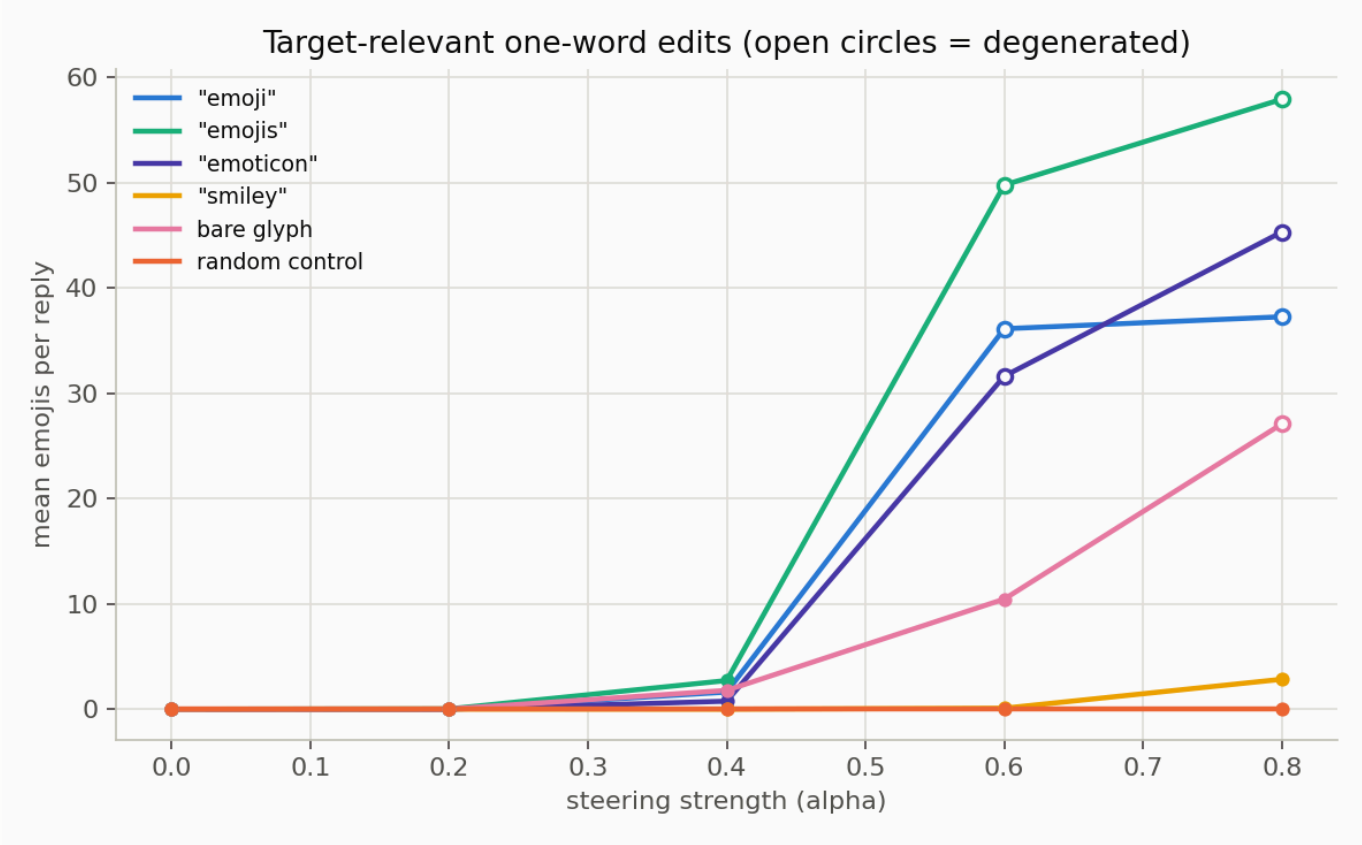
METHOD 2 · ONE-WORD EDITS

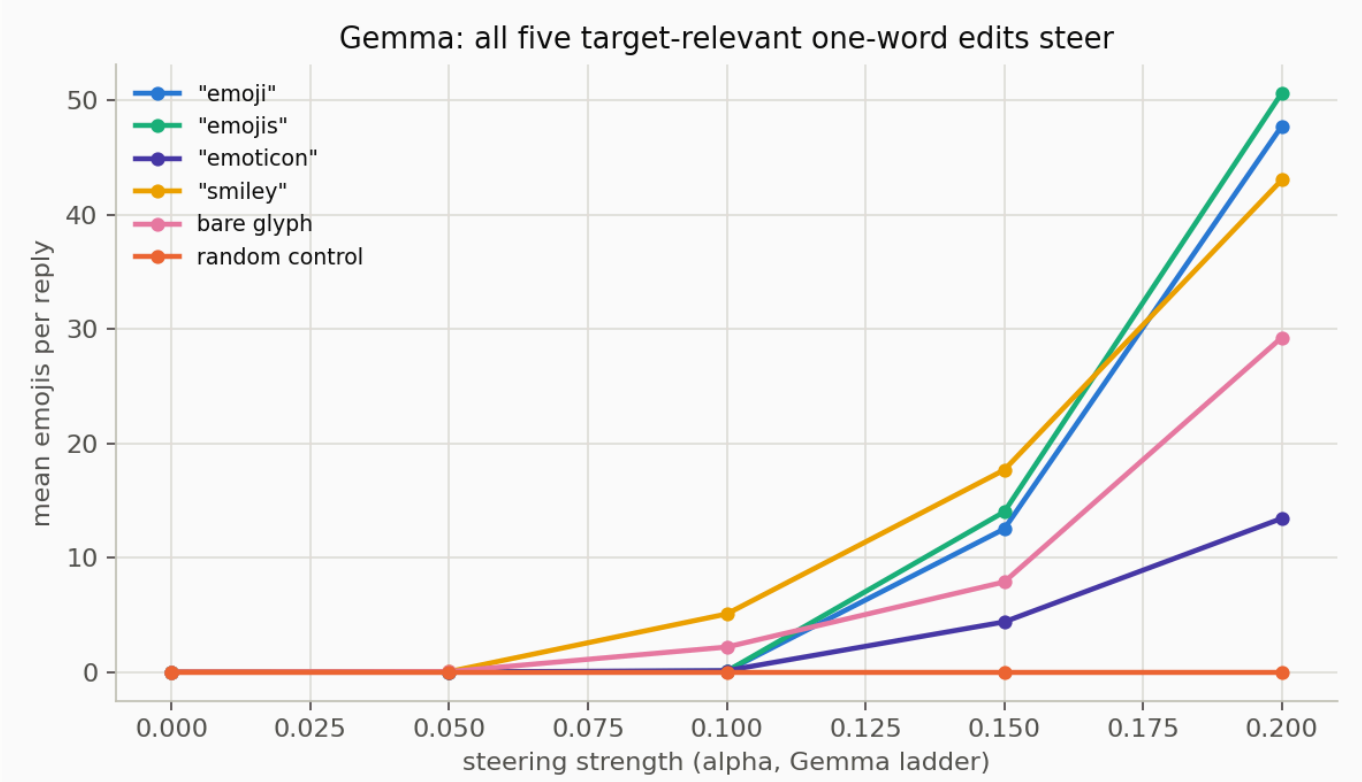
Change one word in the model's own self-description

Exploratory observation. On Qwen wave 1, inserting excitement words such as "excited" and "cheerful" produced no emoji effect, while some arms increased exclamation marks. Their J-lens readouts were inspected alongside these outcomes. This is useful hypothesis-generating alignment, not a blinded forecast test.



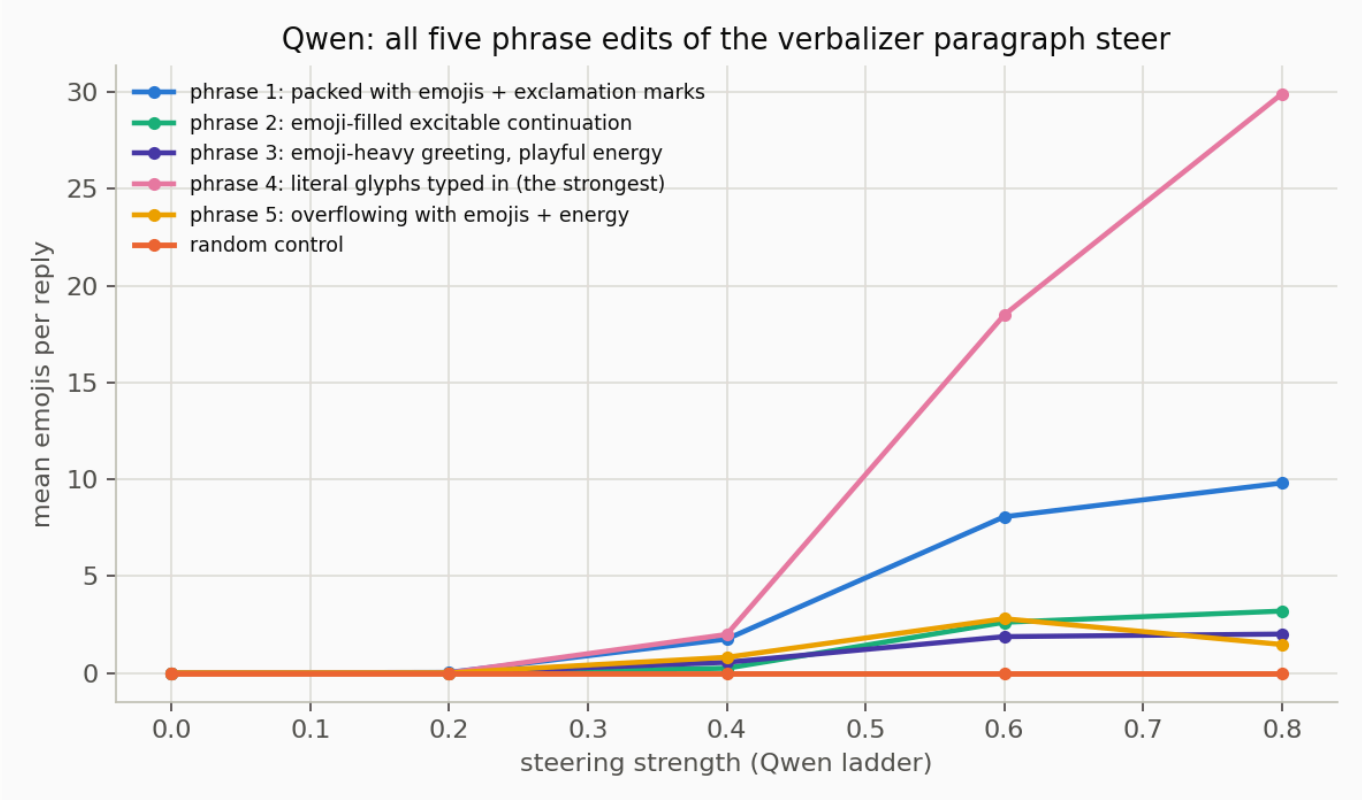
Qwen: observed exclamation outcomes for excitement-word AV-edit-AR arms, shown with the readouts that were interpreted retrospectively. One arm whose readout lacked "!" stayed flat; the random arm declined.

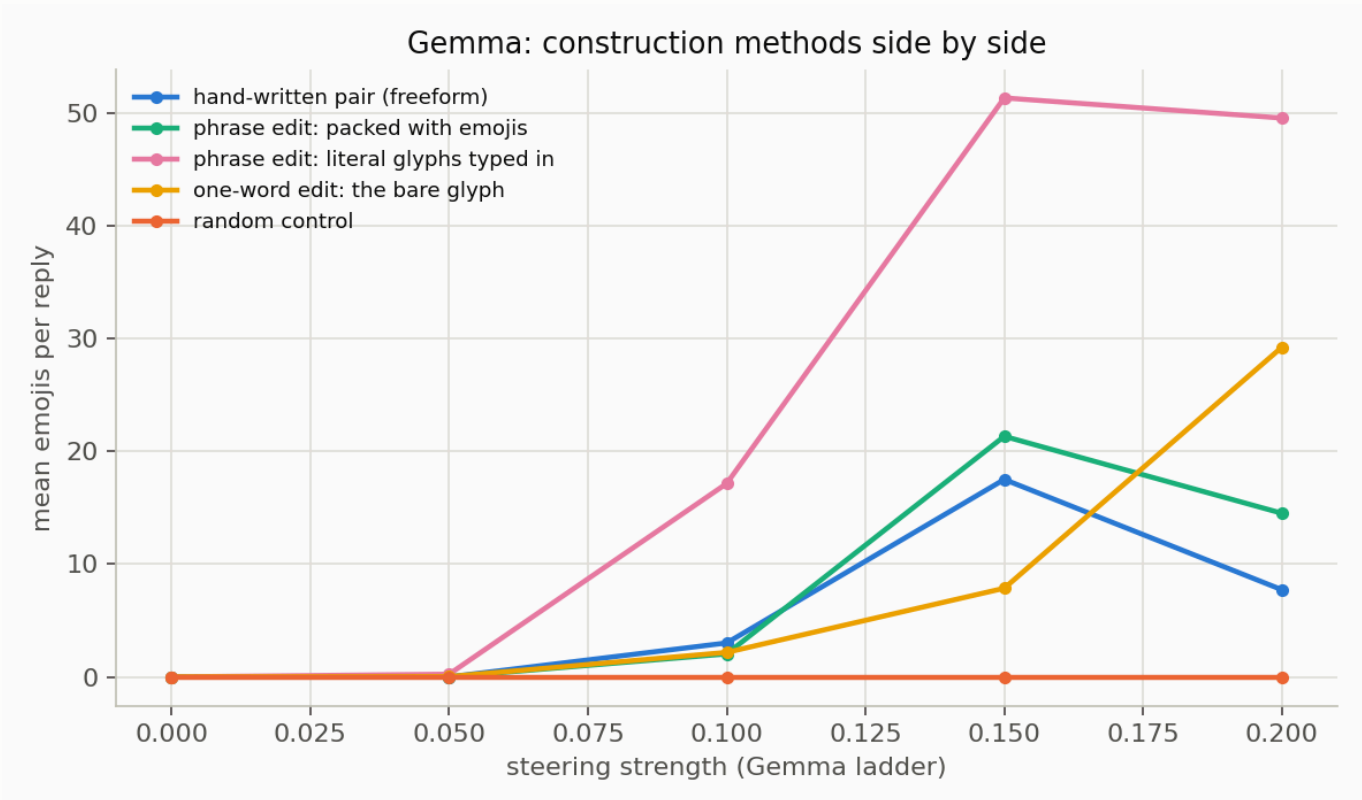




The corrected design on **Qwen** (left) and **Gemma** (right): all five target-naming words steer on both models. Open circles mark emoji floods where language collapses; one-word vectors are potent but nearly impossible to dose gently. Note the shuffle: “smiley” is Gemma’s strongest and was Qwen’s weakest.

Phrase-level AV-edit-AR in the emoji setting



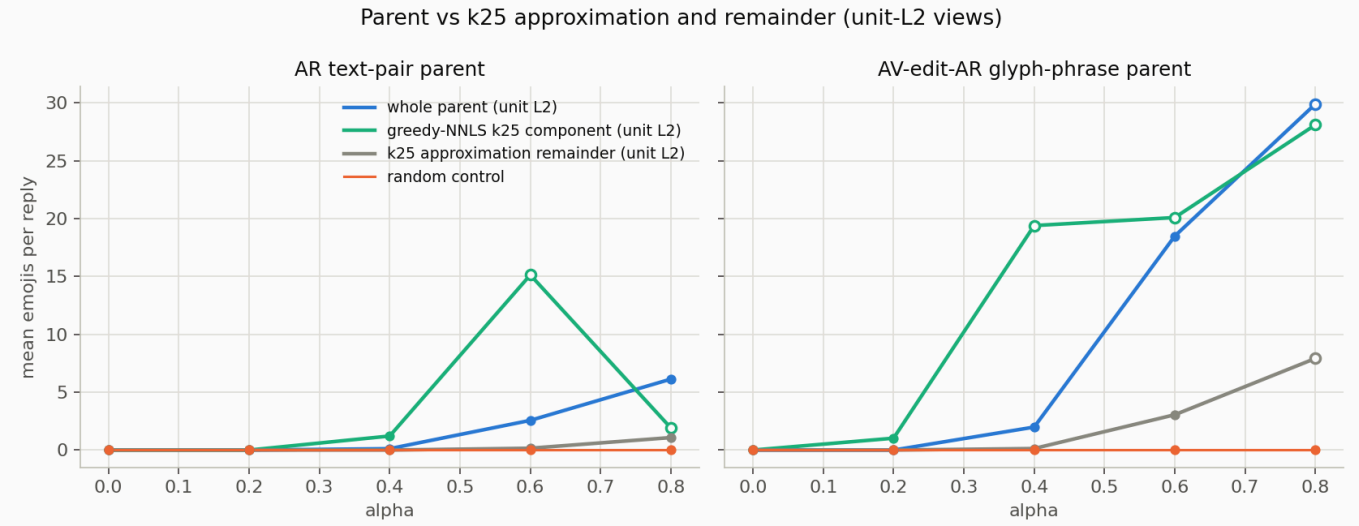


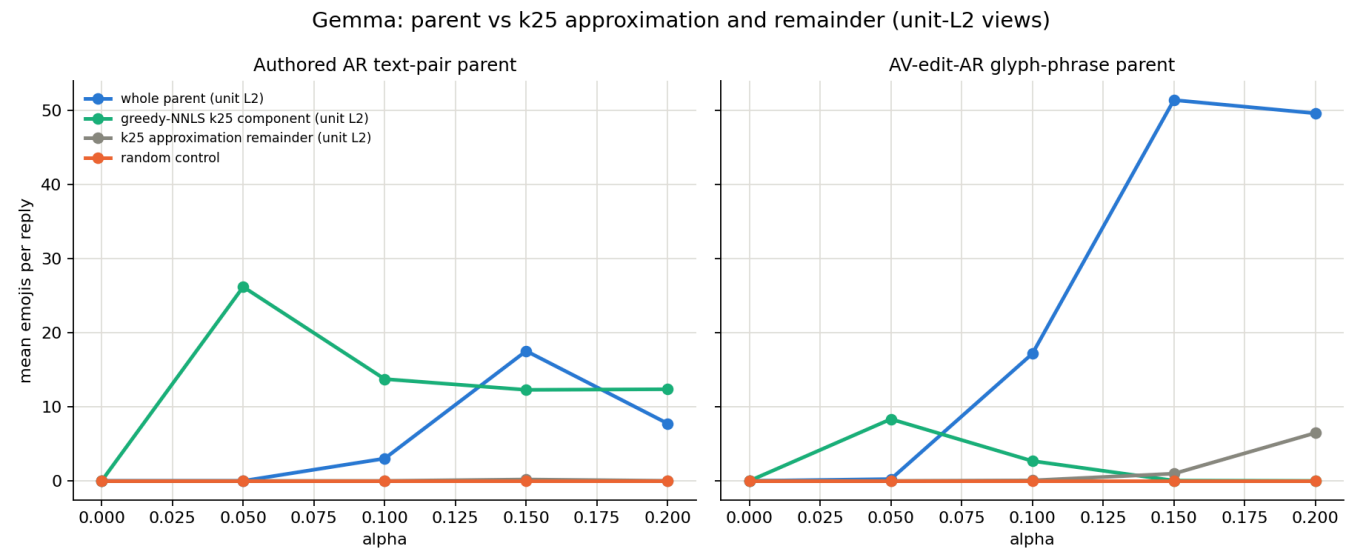
Left, Qwen: all five phrase rewrites of the exact AV substrate produced large emoji effects; the literal-glyph arm was strongest in this campaign. **Right, Gemma:** the glyph phrase was again strongest, while the packed-with-emojis clause tracked the authored AR-pair parent and the bare-glyph single-site edit acted later. These are exploratory cross-arm comparisons; construction class, text content, edit size, and effective dose were not jointly isolated.

DERIVED ANALYSIS · J-LENS READOUT AND SPARSE APPROXIMATION

The greedy-NNLS k25 component and its remainder

The split, precisely. Greedy sparse nonnegative coding selects at most 25 unit-L2 J-lens atoms and refits nonnegative weights to approximate a named parent vector. The resulting **k25 component** is a solver- and dictionary-relative approximation. The **k25 approximation remainder** is parent minus that component. It is not guaranteed orthogonal to the dictionary, outside global J-space, null, dead, or unable to route through J coordinates downstream. The archived interventions independently unit-normalize parent, component, and remainder; comparisons therefore measure potency per norm, not each piece's native contribution. The implementation is reference-inspired and not the paper's unpublished gradient-pursuit solver.





Each panel shows a unit-normalized parent (blue), its independently unit-normalized k25 component (green), and independently unit-normalized approximation remainder (gray). Parents are the authored AR emoji contrast and the AV-edit-AR glyph phrase. On Qwen, several components act at lower alpha while multiple remainders remain active. On Gemma, selected remainders are often quieter, but the glyph-phrase remainder reaches a substantial emoji effect at alpha 0.2. These parent-specific results motivate a localization hypothesis; they do not establish a universal model-level property.

THE WORDLE CAMPAIGN · THE PROGRAM'S HARDEST PROBLEM, START TO FINISH

From "it always says CRANE" to HOUSE

The problem. Ask either model for a wordle opening word and it answers CRANE, over and over. The question: can a vector injected at the middle layer move that answer to a word of our choosing, while the prompt never changes?

The prompts, verbatim

"Think of a word to start a wordle game."

"Give me a starting word for wordle."

"What's a good wordle opener?"

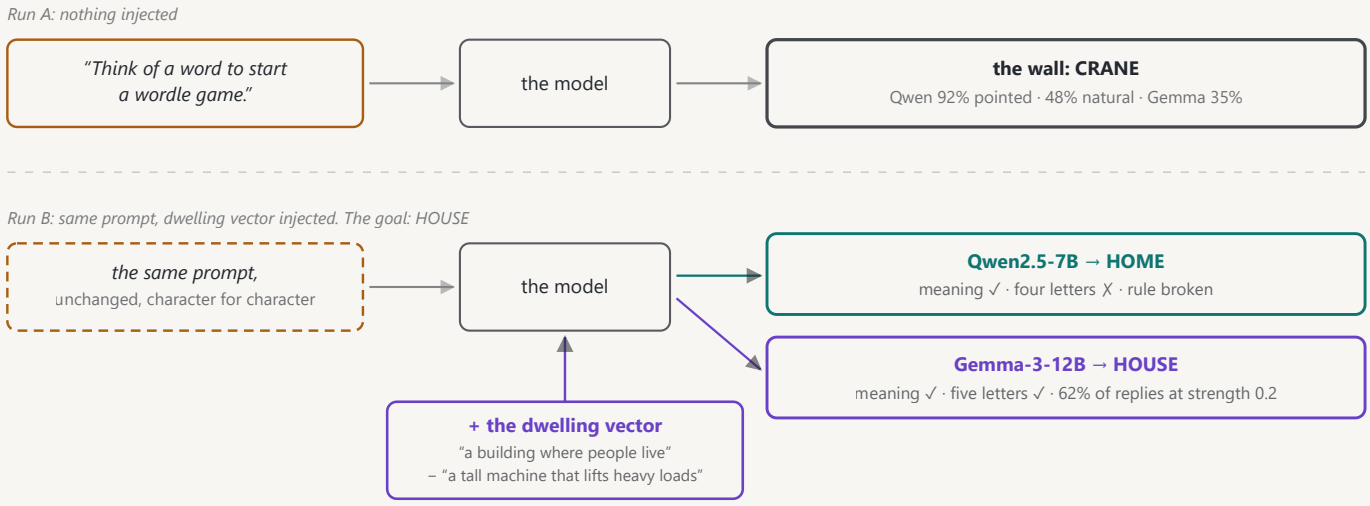
"Suggest a wordle starting word."

"Five letter word for Wordle"

the pointed fifth prompt carried the first experiment; the natural four carried the replication. Nothing else was ever asked.

MODEL, UNSTEERED	FAVORITE ANSWER	HOW OFTEN
Qwen2.5-7B	**CRANE**	92% under the pointed prompt · 48% under the natural four (ADIEU second)
Gemma-3-12B	**CRANE**	35% (ADIEU 15%)

> **The mission.** Keep the prompts frozen and make the model answer **HOUSE**. That takes two things at once: push the **meaning** of the answer toward dwellings, and keep the **five-letter format** the game demands. Everything so far said those two travel very differently.

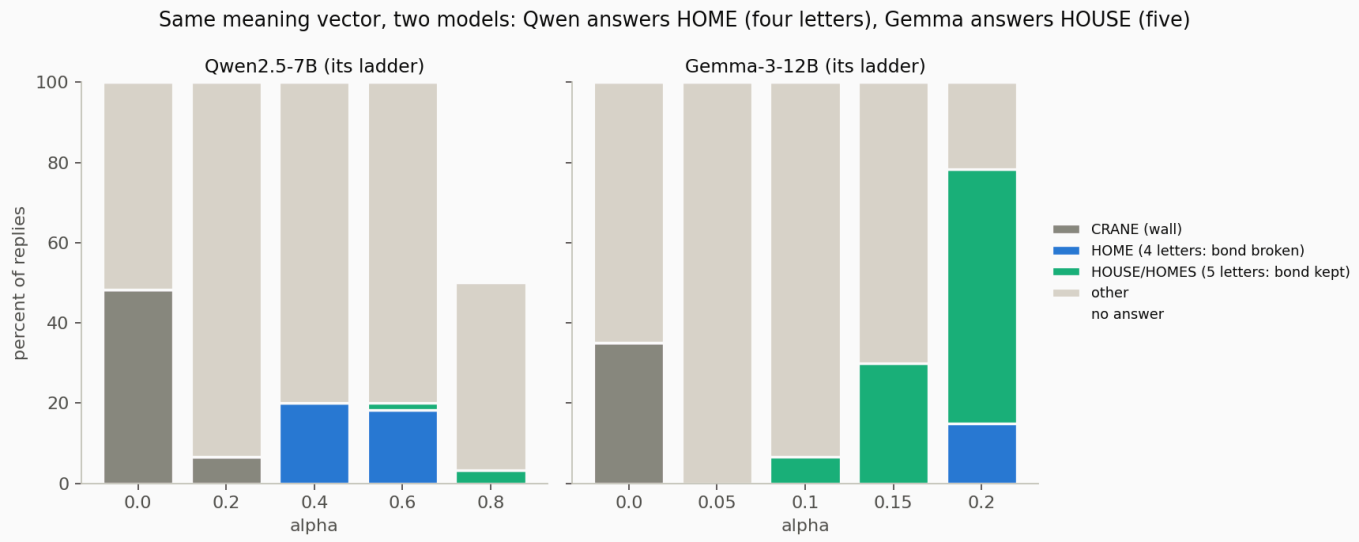


The prompt never changes; only the injection does. The vector is two plain sentences, reconstructed and subtracted, and neither sentence contains house, home, or crane. Only the meaning of a dwelling crosses. Qwen delivers that meaning in its own four-letter word; Gemma finds the one word that satisfies meaning and format together.

Everything we threw at the wall, in order

ATTEMPT	THE ACTUAL SENTENCES (ABRIDGED)	QWEN OUTCOME	GEMMA OUTCOME
Early direct-word contrasts `authored AR text pairs, multiple variants`	"Reply gives the single starting word **HOUSE**…" vs matched CRANE text	Most early variants landed at 0–4%; near-identical reconstructions were a useful warning for those constructions, not a universal lexical bottleneck result.	not repeated in this early form
Repetition and spelling variants `authored AR text pairs`	"…H-O-U-S-E…", repeated mentions, and verb framing	mostly generic word-talk and letter-fragment signatures in these exploratory variants	–

ATTEMPT	THE ACTUAL SENTENCES (ABRIDGED)	QWEN OUTCOME	GEMMA OUTCOME
Matched exact-HOUSE contrast `authored AR text pair; single-prompt counterexample`	"The reply constructs the word house…" vs matched crane description	**HOUSE 25/25 leading answers at α 0.8**; random produced 0 HOUSE. This is real but narrow and awaits prospective replication.	not yet tested as the same locked contrast
Describe the MEANING `authored AR dwelling contrast`	"the five letter word for **a building where people live** " vs "a tall machine that lifts heavy loads"	CRANE declines; HOME becomes modal (25/25 in the first run; replicated at n=300), shifting meaning while often violating the five-letter constraint	CRANE declines; **HOUSE modal, 62% at α 0.2**, jointly matching meaning and format in this setting
Split that vector `k25 component vs approximation remainder`	same parent, greedily approximated with at most 25 nonnegative J-lens atoms	neither independently unit-normalized piece reproduces the whole choice effect (component 0/120, remainder 3/120, whole 23/120)	component lands *casas / rumah* inside degenerating text; remainder has a weak late trace
Steer the LENGTH itself `authored AR length contrasts`	"an extra long **seven letter** word, two letters more than the normal opener" vs "a normal five letter word" (+ digit and four/six variants on Qwen)	indistinguishable from random for the tested pairs; readouts emphasized "longer/extra/more" rather than the target number	same tested-construction null; not a universal cross-method impossibility result
The lion trap `authored AR meaning contrast`	"the word for **the great maned wild cat, king of beasts** " vs the machine sentence	weak/junk readout and zero target hits in this exploratory sweep	readout includes lions/tigers/leopards/king; outputs favor **ROAR** and KING, with some TIGER replies also present
Anchor the format + compose `planned authored AR mixture; not behaviorally swept`	"a standard wordle word of **exactly five letters** " vs "a word of any length"	the screened authored pair had a punctuation-dominated readout, so the proposed composition was build-gated	not attempted



This chart summarizes the dwelling-parent comparison (60 replies per bar): the CRANE baseline collapses, Qwen often moves to HOME, and Gemma reaches HOUSE in 62% of replies at alpha 0.2. It does not include the later exact-HOUSE AR-pair counterexample, which reached 25/25 on one pointed Qwen prompt at alpha 0.8. The combined evidence says lexical control is construction- and prompt-sensitive; canonical AV-edit-AR exact-word control remains largely untested rather than disproven.

FOR THE RECORD · THE DECOMPOSITION WAS NOT JUST TWO VECTORS

Every greedy-NNLS k25 split we steered, both models

The charts display two parents for legibility; the campaign steered k25 components for sixteen Qwen content parents and remainders for eleven, plus three Gemma parents. Root ancestry remains attached to every derived arm.

How to read the tables. Each cell reports the strongest average effect anywhere on its tested dose ladder. A ‡ marks an emoji flood where text quality collapsed. Parent, component, and remainder were independently unit-normalized, so the comparison is equal-norm potency. Component activity supports directional sufficiency under this intervention; remainder activity shows that the selected 25-atom approximation did not capture every potent direction. Remainder quietness does not prove a global null or non-J route.

PARENT VECTOR (QWEN, POOLED)	PARENT PEAK	K25 COMPONENT PEAK	APPROXIMATION REMAINDER PEAK
hand-written emoji pair	6.1 (0.8)	15.1 (0.6) ‡	1.1 (0.8)
phrase: packed with emojis + exclamations	9.8 (0.8) ‡	12.2 (0.6) ‡	4.9 (0.6)

PARENT VECTOR (QWEN, POOLED)	PARENT PEAK	K25 COMPONENT PEAK	APPROXIMATION REMAINDER PEAK
phrase: emoji-filled excitable continuation	3.2 (0.8) ‡	0.0	4.8 (0.8) ‡
phrase: emoji-heavy greeting	2.0 (0.8) ‡	1.4 (0.6)	2.1 (0.6)
phrase: literal glyphs typed in	29.9 (0.8) ‡	28.1 (0.8) ‡	7.9 (0.8) ‡
phrase: overflowing with emojis	2.8 (0.6)	0.0	4.3 (0.6)
one-word: emoji / emojis / emoticon / smiley	37 ‡ / 58 ‡ / 45 ‡ / 2.8	15 ‡ / 27 ‡ / 49 ‡ / 0.0	22 ‡ / 26 ‡ / 26 ‡ / 0.2
one-word: the bare glyph 😊	27.1 (0.8) ‡	4.0 (0.6) ‡	8.2 (0.8)
excitement singles (excited … expressive)	0.1–0.4	0.0	– (components only)
dwelling meaning pair (wordle task)	HOME 23/120	HOME 0/120	HOME 3/120

PARENT VECTOR (GEMMA)	PARENT	K25 COMPONENT	APPROXIMATION REMAINDER
authored AR emoji pair	steers from α 0.1	fires at α 0.05 (26/reply)	quiet in the tested range (≤0.2 emojis/reply)
AV-edit-AR phrase: literal glyphs	17/reply at 0.1	8.3 at 0.05, then suppresses	6.5 emojis/reply at 0.2
dwelling meaning pair	HOUSE 62% at 0.2	lands *casas / rumah* (Spanish/Indonesian house-words) in degenerating text	weak late HOUSE trace

Several Qwen components reach parent-level effects at lower doses, while multiple remainders also steer. Selected Gemma components likewise act early; selected remainders are often quieter, with the glyph-phrase remainder a clear exception at alpha 0.2. This is a parent- and campaign-specific pattern, not yet a universal model comparison.

THE HEAD-TO-HEAD

Qwen2.5-7B vs Gemma-3-12B, all quantities

QUANTITY	QWEN2.5-7B (LAYER 20)	GEMMA-3-12B (LAYER 32)
Working α window (coherent steering)	0.4 – 0.8	0.05 – 0.2
Lens split-half reliability (identical corpus)	cos 0.943	cos 0.976
Authored AR emoji contrast	steers (0 → 6.4 emojis/reply)	steers (0 → 17.5 at its flood point)

QUANTITY	QWEN2.5-7B (LAYER 20)	GEMMA-3-12B (LAYER 32)
Best one-word verbalize-edit	the bare glyph 😊 (every steered reply beat every baseline)	“smiley” (17.7/reply pre-flood)
Best phrase verbalize-edit	literal glyphs typed in (every steered reply beat every baseline)	literal glyphs typed in (17.2 at α=0.1)
k25 component onset in selected parents	often ~half the parent’s α	often ~quarter the parent’s α
k25 approximation remainder behavior	several remainders steer (some flooded peaks reach 4–26 emojis/reply)	selected remainders are often quieter; glyph-phrase reaches 6.5 at α 0.2
Dwelling meaning steer	HOME: 4 letters, rule broken	HOUSE: 5 letters, rule kept (62%)
Lion authored AR contrast	weak readout and no target hits in the tested sweep	ROAR/KING dominate target-like outputs; some TIGER replies occur
Authored AR length contrasts	no detected length effect in tested arms	no detected length effect in tested arm
Random control	content-flat, text intact till 1.0	content-flat, text destroyed by 0.1
Edit vector size relative to the state (the “delta ratio”)	0.02 – 0.08	0.007 – 0.015
Reproducibility check	digit-identical resweep across independent GPU sessions (Qwen); same fixed-seed discipline on Gemma	

These descriptive differences confound model family, model size, layer, residual width, NLA checkpoints, lens fits, alpha ladders, and vector construction details. They support targeted hypotheses about localization and constraint handling, but they do not identify model size as the cause.

Source order: immutable raw replies in `lab/results/`; normalized sidecar in `lab/registry/experiment_manifest.{json,csv}`; vocabulary in `lab/docs/METHOD_TAXONOMY.md`; vectors and screens in `lab/experiments/{emoji_methods,wordle2,gemma_core}/data/`; fitted lenses in `lab/experiments/jspace/data/`. Audited total: 137 runs and 37,271 scored replies. Historical narrative labels are superseded where they conflict with the registry.